

In the format provided by the authors and unedited.

Early prediction of circulatory failure in the intensive care unit using machine learning

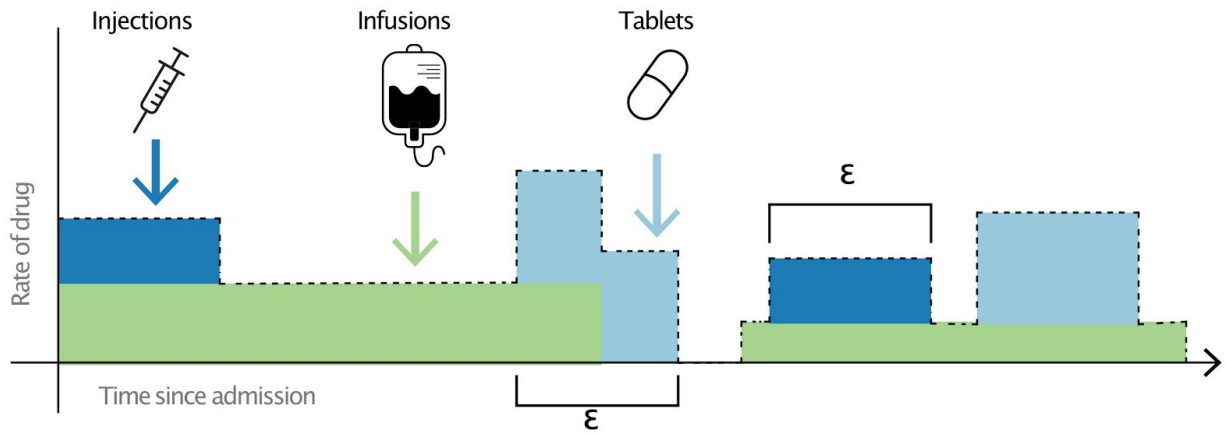
Stephanie L. Hyland^{1,2,3,4,10}, Martin Faltys^{5,10}, Matthias Hüser^{1,4,10} , Xinrui Lyu^{1,4,10}, Thomas Gumbsch^{6,7,10}, Cristóbal Esteban^{1,4}, Christian Bock^{1,6,7} , Max Horn^{6,7}, Michael Moor^{6,7}, Bastian Rieck^{6,7}, Marc Zimmermann¹, Dean Bodenham^{6,7}, Karsten Borgwardt^{1,6,7,11}  , Gunnar Rätsch^{1,2,3,4,7,8,11}   and Tobias M. Merz^{1,5,9,11}  

¹Department of Computer Science, ETH Zürich, Zürich, Switzerland. ²Computational Biology Program, Memorial Sloan Kettering Cancer Center, New York, NY, USA. ³Tri-Institutional PhD Program in Computational Biology and Medicine, Weill Cornell Medicine, New York, NY, USA. ⁴Medical Informatics Unit, Zürich University Hospital, Zürich, Switzerland. ⁵Department of Intensive Care Medicine, University Hospital, University of Bern, Bern, Switzerland. ⁶Department of Biosystems Science and Engineering, ETH Zürich, Basel, Switzerland. ⁷Swiss Institute for Bioinformatics, Lausanne, Switzerland. ⁸Department of Biology, ETH Zürich, Zürich, Switzerland. ⁹Cardiovascular Intensive Care Unit, Auckland City Hospital, Auckland, New Zealand. ¹⁰These authors contributed equally: Stephanie L. Hyland, Martin Faltys, Matthias Hüser, Xinrui Lyu, Thomas Gumbsch. ¹¹These authors jointly supervised this work: Karsten Borgwardt, Gunnar Rätsch, Tobias M. Merz. ✉e-mail: karsten.borgwardt@bsse.ethz.ch; gunnar.raetsch@inf.ethz.ch; tobiasm@adhb.govt.nz

Supplementary Information to:

Early prediction of circulatory failure in the intensive care unit using machine learning

Stephanie L. Hyland, Martin Faltys, Matthias Hüser, Xinrui Lyu, Thomas Gumbsch, Cristóbal Esteban, Christian Bock, Max Horn, Michael Moor, Bastian Rieck, Marc Zimmermann, Dean Bodenham, Karsten Borgwardt, Gunnar Rätsch, Tobias M. Merz



Supplementary Figure 1: Processing details for pharmaceutical variables. An effective flow rate of a drug was calculated by merging continuous infusions, intravenous boluses, and tablets. Intravenous bolus applications and tablets do not result in a constant effective blood drug level and were therefore converted to an effective rate by dividing the given dose by the acting period of the drug. These acting periods were specified using clinical domain knowledge and are listed in Supplementary Table 4.

Supplementary Table 1: Multiscale feature parameters

Frequency	Median sampling interval	Short horizon	Medium horizon	Long horizon	Very long horizon
High	< 15 mins	30 mins	60 mins	240 mins	720 mins
Medium	>15 mins, < 8 hours	12 hours	24 hours	36 hours	48 hours
Low	>= 8 hours	16 hours	32 hours	48 hours	72 hours

Overview of multi-scale horizon lengths used for the different clinical variables during feature construction, conditional on the length of a typical sampling interval of a variable, shown in the second column. Variables were binned into frequency categories high, medium, and low. For non-pharmaceutical variables, the sampling interval was computed from the intervals between observations, and for drugs, it corresponded to the effective acting periods compiled using clinical knowledge.

Supplementary Table 2: List of instability features

Feature type	Conditions	Horizon lengths
Fraction of time in sub-event	[MAP $\leq$ 65 mmHg], [vLac $\geq$ 2 or aLac $\geq$ 2 mmol/l], [Dop $>$ 0 μg/min], [Mil $>$ 0 μg/min], [Lev $>$ 0 μg/min], [Theo $>$ 0 μg/min], Event L1 := [(vLac $\geq$ 2 or aLac $\geq$ 2 mmol/l) and (MAP $\leq$ 65 mmHg or Dop $>$ 0 μg/min or Mil $>$ 0 μg/min or Theo $>$ 0 μg/min or Lev $>$ 0 μg/min)], [Noreph $>$ 0 μg/min and Noreph $<$ 0.1 μg/min], [Noreph $\geq$ 0.1 μg/min], [Epineph $>$ 0 μg/min and Epineph $<$ 0.1 μg/min], [Epineph $\geq$ 0.1 μg/min] Event L2 := [(vLac $\geq$ 2 or aLac $\geq$ 2 mmol/l) and (Noreph $>$ 0 μg/min and Noreph $<$ 0.1 μg/min or Epineph $>$ 0 μg/min and Epineph $<$ 0.1 μg/min)], [Vaso $>$ 0 μg/min], Event L3 := [(vLac $\geq$ 2 or aLac $\geq$ 2 mmol/l) and (Noreph $>$ 0.1 μg/min or Epineph $>$ 0.1 μg/min or Vaso $>$ 0 μg/min)]	12 hours, 24 hours, 36 hours, 48 hours, Entire stay
Time to last occurrence of sub-event (if any); otherwise set to a large value equal to 30 days if there was no occurrence of event	Same as first row	N/A
Current sub-event state	Same as first row	N/A

Specification of the instability features computed from specific variables relevant for circulatory failure. Three subtypes of features were constructed, corresponding to rows. The exact specification of individual conditions is listed in brackets in the second column. The last column, in analogy to the multi-scale summaries, displays the horizon lengths over which time-fraction features are computed. Non-adaptive constant horizons were used. For each condition in the first row, time-to-last occurrence and current event state features were defined similarly. Norepinephrine is abbreviated as noreph. N/A: Not applicable. MAP: Mean arterial pressure, vLac: Venous lactate, aLac: Arterial lactate, Dop: Dobutamine, Mil: Milrinone, Lev: Levosimendan, Theo: Theophyllin, Noreph: Norephenephine, Epineph: Epinephrine, Vaso: Vasopressin.

Supplementary Table 3: Shapelet extraction details

Frequency	Median sampling interval	Search window	Min shapelet length	Max shapelet length
High	< 15 mins	240 mins	30 mins	60 mins
Medium	< 15 mins, < 8 hours	36 hours	12 hours	24 hours
Low	>= 8 hours	48 hours	16 hours	32 hours

The shapelet extraction was performed on time series subsequences that precede onset of circulatory failure where the search window and the size of the shapelet depends on the variable sampling frequency, which is displayed in the table.

Supplementary Table 4: List of variables in *circEWS* model

This table lists the full set of variables used in our models. The “full” model uses 112 variables associated with the top 500 features. The “compact” model uses only features from the top-20 variables, excluding those not identifiable in MIMIC, resulting in 16 variables generating 176 features. Where relevant, we further provided the chosen mapping to MIMIC ITEMIDs, as well as the default values used in imputation (selected using prior clinical knowledge), and the range of permitted values for artefact removal. Additional sheets detailing variable/drug merging heuristics and effective imputation ratios of measured variables were also included. Please refer to the attached Excel file <circEWS_Variables_Overview_Table.xlsx>.

Columns in this spreadsheet

- Variable name
- #features derived from this variable
- Inclusion of variable in compact model
- Relevant MIMIC identifiers
- Default value
- Permitted range of values
- Type (continuous, binary, categorical)

Further sheets:

- List of drugs comprising combined drugs
- List of variables formed from merging other variables (non-drugs)
- Effective imputation ratios of measured variables

Supplementary Table 5: Model calibration in patient subgroups

	Full model				Compact model		
	Cohort size	Raw score	Cal. score	Raw score (time)	Raw score	Cal. score	Raw score (time)
Sex							
Male	1446 (164)	0.38 (0.01)	0.04 (0.02)	0.21 (0.10)	0.39 (0.01)	0.04 (0.01)	0.15 (0.08)
Female	863 (93)	0.39 (0.01)	0.05 (0.01)	0.34 (0.10)	0.39 (0.01)	0.06 (0.02)	0.27 (0.15)
Surgical status							
Non-surgical	1307 (167)	0.38 (0.01)	0.04 (0.02)	0.23 (0.09)	0.39 (0.01)	0.05 (0.01)	0.19 (0.08)
Surgical	1011 (125)	0.39 (0.01)	0.04 (0.02)	0.24 (0.02)	0.40 (0.01)	0.04 (0.02)	0.20 (0.17)
Emergency status							
Emergency	1239 (128)	0.38 (0.01)	0.04 (0.02)	0.23 (0.09)	0.39 (0.01)	0.05 (0.01)	0.19 (0.08)
Elective	1011 (125)	0.39 (0.01)	0.04 (0.01)	0.24 (0.20)	0.41 (0.01)	0.05 (0.02)	0.23 (0.20)
Age							
16-30*	90 (12)	0.39 (0.03)	0.17 (0.09)	1.12 (0.22)	0.41 (0.01)	0.17 (0.10)	0.95 (0.26)
31-45	203 (27)	0.37 (0.02)	0.07 (0.02)	0.69 (1.08)	0.40 (0.01)	0.07 (0.02)	0.98 (1.06)
46-65	820 (99)	0.38 (0.01)	0.05 (0.01)	0.45 (0.07)	0.39 (0.01)	0.04 (0.01)	0.46 (0.07)
66-80	932 (109)	0.39 (0.01)	0.04 (0.01)	0.24 (0.11)	0.39 (0.01)	0.04 (0.01)	0.16 (0.05)
81-100	265 (34)	0.40 (0.01)	0.08 (0.02)	0.34 (0.41)	0.40 (0.02)	0.09 (0.04)	0.36 (0.29)
APACHE score							
0-15	1059 (192)	0.41 (0.01)	0.07 (0.02)	0.40 (0.95)	0.42 (0.01)	0.08 (0.03)	0.41 (0.81)
16-25	771 (82)	0.38 (0.01)	0.04 (0.01)	0.43 (0.12)	0.39 (0.01)	0.05	0.40 (0.10)

						(0.01)	
26-35	268 (42)	0.37 (0.02)	0.06 (0.03)	0.54 (0.12)	0.38 (0.02)	0.07 (0.04)	0.56 (0.11)
36-45*	41 (14)	0.36 (0.04)	0.11 (0.03)	0.52 (0.35)	0.36 (0.04)	0.10 (0.03)	0.39 (0.36)
46-57*	4 (2)	0.40 (0.07)	0.30 (0.15)	1.31 (0.19)	0.39 (0.04)	0.29 (0.13)	1.04 (0.21)
Patient group							
Cardiovascular	762 (114)	0.39 (0.01)	0.05 (0.02)	0.57 (0.06)	0.40 (0.01)	0.05 (0.02)	0.51 (0.04)
Neurological	686 (60)	0.40 (0.01)	0.15 (0.11)	0.38 (0.10)	0.41 (0.02)	0.15 (0.12)	0.42 (0.12)
Gastrointestinal	357 (58)	0.37 (0.01)	0.06 (0.01)	0.22 (0.13)	0.38 (0.01)	0.06 (0.03)	0.16 (0.22)
Respiratory	197 (24)	0.37 (0.02)	0.09 (0.03)	0.65 (0.20)	0.38 (0.01)	0.09 (0.06)	0.67 (0.14)
Other	135 (24)	0.40 (0.02)	0.12 (0.05)	1.11 (0.78)	0.41 (0.02)	0.11 (0.05)	0.86 (0.91)
Trauma	111 (16)	0.39 (0.02)	0.10 (0.07)	0.64 (0.19)	0.41 (0.02)	0.12 (0.07)	0.61 (0.22)
Metabolic/Endocrinology	46 (13)	0.37 (0.02)	0.11 (0.04)	0.41 (0.16)	0.38 (0.03)	0.12 (0.04)	0.61 (0.22)

Model calibration in patient subgroups of the HiRID cohort. 'Raw score' and 'Cal. score.' denote the area between the calibration curve and the ideal diagonal curve for the raw scores and calibrated scores (isotonic regression), respectively. 'Raw score (time)' denotes the mean absolute difference in hours to a linear fit to the mean values of the raw prediction score against time to deterioration. Patient groups with at most 100 patients per split on average are marked with * and are likely affected by statistical estimation errors. Numbers in parentheses indicate the standard deviation across n=6 independent experiments in the temporal splits.

Supplementary Table 6: Effect of silencing period

Silencing period [min]	AUPRC	Prec@80% Recall	Prec@90% Recall	Prec@95% Recall
30	0.63 (0.027)	0.441 (0.024)	0.296 (0.024)	0.164 (0.028)
60	0.633 (0.024)	0.446 (0.026)	0.298 (0.025)	0.166 (0.024)
90	0.635 (0.023)	0.444 (0.025)	0.299 (0.023)	0.169 (0.033)
120	0.641 (0.020)	0.45 (0.025)	0.302 (0.022)	0.162 (0.047)
150	0.644 (0.020)	0.458 (0.025)	0.286 (0.019)	0.161 (0.046)
180	0.646 (0.021)	0.454 (0.022)	0.289 (0.019)	0.164 (0.046)
210	0.644 (0.020)	0.461 (0.021)	0.283 (0.021)	0.15 (0.055)
240	0.645 (0.018)	0.464 (0.020)	0.288 (0.021)	0.13 (0.05)
270	0.644 (0.017)	0.455 (0.018)	0.287 (0.016)	0.132 (0.05)
300	0.643 (0.017)	0.443 (0.018)	0.263 (0.014)	0.13 (0.051)
330	0.646 (0.017)	0.453 (0.019)	0.284 (0.017)	0.132 (0.042)
360	0.645 (0.017)	0.443 (0.015)	0.278 (0.013)	0.131 (0.027)
390	0.639 (0.017)	0.445 (0.014)	0.252 (0.017)	0.049 (0.004)
420	0.634 (0.017)	0.44 (0.014)	0.247 (0.021)	0.05 (0.005)
450	0.631 (0.018)	0.441 (0.016)	0.239 (0.020)	0.051 (0.004)

We fixed a silencing window of 30 minutes. Here we show how overall performance (assessed by AUPRC and precision at fixed recall) varies as we modify this silencing time. We see that up to 7 hours of silencing results in diminished, but still high performance. The long silencing time is mitigated by the unsilencing mechanic, whereby a patient entering and recovering from a period of circulatory failure will cause silencing to be terminated early. Numbers in parentheses indicate the standard deviation across n=6 independent experiments performed per temporal split. Prec: Precision, AUPRC: Area under the precision recall curve.

Supplementary Table 7: Effect of reset time

System reset time [min]	AUPRC	Prec@80% Recall	Prec@90% Recall	Prec@95% Recall
0	0.668 (0.021)	0.497 (0.028)	0.367 (0.028)	0.248 (0.023)
5	0.668 (0.021)	0.497 (0.028)	0.367 (0.028)	0.248 (0.023)
10	0.658 (0.022)	0.481 (0.025)	0.35 (0.027)	0.225 (0.028)
15	0.617 (0.025)	0.464 (0.027)	0.334 (0.029)	0.207 (0.035)
20	0.612 (0.026)	0.45 (0.026)	0.311 (0.023)	0.19 (0.036)
25	0.606 (0.031)	0.445 (0.028)	0.296 (0.024)	0.17 (0.032)
30	0.6 (0.031)	0.425 (0.022)	0.266 (0.020)	0.139 (0.032)
35	0.589 (0.030)	0.405 (0.018)	0.257 (0.026)	0.128 (0.016)
40	0.582 (0.031)	0.394 (0.018)	0.227 (0.028)	0.074 (0.012)
45	0.57 (0.031)	0.377 (0.028)	0.19 (0.036)	0.04 (0.011)
50	0.552 (0.031)	0.343 (0.026)	0.131 (0.043)	0.04 (0.011)

We study the impact of silencing the alarm after a patient exited from a state of circulatory failure, the reset time. Increasing the reset time results in lower AUPRC and precision due to missed deterioration events. The default value for reset time in *circEWS* is 22.5 minutes. Numbers in parentheses indicate the standard deviation across $n=6$ independent experiments performed per temporal split. Prec: Precision, AUPRC: Area under the precision recall curve.

Supplementary Table 8: Effect of grey zone

Grey zone before evaluation window / after event [h]	Prec@80% Recall	Prec@90% Recall	Prec@95% Recall
-8	0.506 (0.167)	0.395 (0.098)	0.170 (0.093)
-7	0.505 (0.167)	0.393 (0.098)	0.169 (0.093)
-6	0.505 (0.167)	0.394 (0.098)	0.169 (0.093)
-5	0.505 (0.167)	0.393 (0.098)	0.168 (0.093)
-4	0.505 (0.167)	0.393 (0.098)	0.168 (0.093)
-3	0.504 (0.168)	0.393 (0.100)	0.167 (0.093)
-2	0.503 (0.168)	0.392 (0.100)	0.167 (0.092)
-1	0.502 (0.170)	0.391 (0.102)	0.167 (0.092)
0	0.500 (0.174)	0.390 (0.102)	0.166 (0.092)
1	0.502 (0.173)	0.394 (0.103)	0.170 (0.092)
2	0.503 (0.170)	0.396 (0.102)	0.173 (0.093)
3	0.504 (0.168)	0.398 (0.101)	0.174 (0.093)
4	0.505 (0.168)	0.399 (0.101)	0.174 (0.093)
5	0.505 (0.168)	0.400 (0.101)	0.175 (0.093)
6	0.505 (0.167)	0.401 (0.101)	0.176 (0.094)
7	0.505 (0.167)	0.402 (0.101)	0.177 (0.094)
8	0.506 (0.167)	0.403 (0.101)	0.178 (0.094)

We studied the effect of a grey zone where alarms are not counted as false positives. A grey zone of one hour did not lead to significantly larger precision in any direction at a p-value threshold of 0.05 using a student two-sample *t*-test. Numbers in parentheses indicate the standard deviation across n=6 independent experiments per temporal split. Prec: Precision.

Supplementary Table 9: Reclassification analysis

Cases	FNR circEWS	TPR circEWS
FNR Baseline	0.14 (0.11)	0.57 (0.18)
TPR Baseline	0.04 (0.13)	0.25 (0.06)
Controls	TNR circEWS	FPR circEWS
TNR Baseline	0.79 (0.06)	0.08 (0.18)
FPR Baseline	0.10 (0.17)	0.03 (0.04)

The fraction of reclassified timepoints with the standard deviation from the cell values across temporal splits. Comparing the off diagonal elements for the cases we note that they are outside of each others standard deviation and thus a model difference between the Baseline and *circEWS* can be assumed. Formally, the net reclassification index describes the fraction of reclassified timepoints. For cases, $NRI_e = 0.14$, for controls, $NRI_{ne} = -0.03$ that add to $NRI=0.11$, again showing that the Baseline and *circEWS* classify time-points differently. Numbers in parentheses indicate the standard deviation across $n=6$ independent experiments in the temporal split. FNR: False negative rate, TPR: True positive rate, FPR: False positive rate, TNR: True negative rate.

Supplementary Table 10: Alarm system baseline thresholds

Variable name	Abnormal value range
Lactate	> 2.2 mmol/L
MAP	< 65 mmHg OR > 150 mmHg
Heart rate	< 60 OR > 110
Cardiac output	< 4 L/min
RASS	< - 2 OR $\geq$ 2
INR	> 1.2
Serum glucose	< 4.0 mmol/L OR > 7.8 mmol/L
C-reactive protein	> 10.0 mg/L
Diastolic BP	< 40 mmHg OR > 140 mmHg
Peak inspiratory pressure (Ventilator)	> 30 cmH ₂ O
Systolic BP	< 90 mmHg OR > 200 mmHg
SpO ₂	< 90%

We consider a simple alarm system that raises an alarm whenever any abnormal value is observed for one of the *circEWS-lite* variables (excluding static and pharmaceutical variables). The abnormal value ranges are listed above. MAP: Mean arterial pressure. RASS: Richmond agitation-sedation scale, INR: International normalized ratio, BP: Blood pressure, SpO₂: Peripheral oxygen saturation.

Supplementary Table 11: Forward variable selection

Forward variable selection rank (std)	Variable	Marginal AUPRC improvement
1 (0.0)	Lactate	0.29
2 (1.2)	MAP	0.109
3 (7.0)	Non-opioid analgesics	0.028
4 (4.3)	Peak pressure (Ventilator)	0.006
5 (5.2)	Supplemental oxygen	0.009
6 (3.5)	Time since ICU admission	0.001
7-10 (3.6, 1.9, 1.9, 3.2)	Dobutamine, Milrinone, Levosimendan, Theophylline	None
11 (4.8)	C-reactive protein	None
12 (5.6)	RASS	None
13 (3.0)	Cardiac output	0.003
14 (2.4)	Serum glucose	None
15 (7.1)	Diastolic BP	None
16 (3.8)	SpO2	None
17 (5.6)	Heart rate	None
18* (3.2)	Patient age	0.004
19 (5.0)	INR	None
20 (6.0)	Systolic BP	None

Shown is the rank (column 1) of a variable (column 2) and performance increase (column 3) in forward variable selection. Point estimates of the ranks were obtained on the held-out split and the standard deviations of the ranks were obtained by running the same procedure on the five development splits, indicated in parentheses. The performance was measured using AUPRC (time-slice based evaluation) on the validation set. A lightGBM model with fixed hyperparameters was used (at most 64 leaves per tree, and column/row sampling ratios of 75% per tree, using early stopping after 20 epochs of no change). The third column indicates the AUPRC improvement obtained by adding this variable to the best model obtained so far, only using variables in rows above the variable. 'None' indicates that no improvement resulted from the addition of the variable. The best model contains 18 variables (and the last variable included in the model is marked with *). AUPRC: Area under the precision-recall curve, MAP: Mean arterial pressure, RASS: Richmond agitation-sedation scale, BP: Blood pressure, SpO2: Peripheral oxygen saturation, INR: International normalized ratio.

Supplementary Table 12: Feature ablation by class

Feature type set	AUPRC	prec (rec=0.8)	prec (rec=0.9)	prec (rec=0.95)
all	0.637 ± 0.027	0.441 ± 0.024	0.296 ± 0.024	0.164 ± 0.028
-static	0.630 ± 0.027	0.422 ± 0.031	0.300 ± 0.031	0.168 ± 0.027
-measurement intensity	0.606 ± 0.030	0.413 ± 0.026	0.286 ± 0.028	0.156 ± 0.022
-multiresolution summaries	0.606 ± 0.021	0.387 ± 0.026	0.261 ± 0.026	0.151 ± 0.034
-instability history	0.634 ± 0.030	0.422 ± 0.022	0.288 ± 0.026	0.178 ± 0.027
-time to instability	0.631 ± 0.026	0.443 ± 0.026	0.303 ± 0.025	0.177 ± 0.019
-shapelets	0.632 ± 0.035	0.434 ± 0.032	0.268 ± 0.032	0.181 ± 0.028
-entire stay horizon	0.635 ± 0.028	0.429 ± 0.020	0.291 ± 0.025	0.180 ± 0.023
only current values	0.541 ± 0.020	0.372 ± 0.036	0.255 ± 0.037	0.128 ± 0.026
only shapelets	0.447 ± 0.063	0.265 ± 0.031	0.201 ± 0.013	0.131 ± 0.005

Comparison of model performance when the reference model is ablated by removing individual feature categories. In addition, two baseline models were used. One uses only the ‘current’ (estimated) values for all variables and includes no special features summarizing the patient history. ‘shapelets’ used the shapelet mining technique to construct features predictive of circulatory failure. +/- indicates standard deviation across n=5 independent experiments in the temporal split, the point estimate refers to the performance in the held-out split. prec: Precision, rec: Recall, AUPRC: Area under the precision-recall curve.

Supplementary Table 13: Standard intervals

Variable name	Baseline interval
Lactate	8 h
MAP	2 min
Heart rate	2 min
Cardiac output	5 min
RASS	1 h
INR	24 h
Serum glucose	24 h
C-reactive protein	24 h
Diastolic BP	2 min
Peak inspiratory pressure (Ventilator)	2 min
Systolic BP	2 min
SpO2	2 min

Standard intervals for non-static variables used in the resampling process (details described in online methods) for the “bias by intensity of monitoring” experiment. These are clinically specified baseline frequencies provided by the clinicians from the Inselspital in Bern. MAP: Mean arterial pressure, RASS: Richmond agitation-sedation scale, INR: International normalized ratio, BP: Blood pressure, SpO2: Peripheral oxygen saturation.

Supplementary Table 14: Description of time-point artefacts and corresponding artifact correction rules in HiRID

Timestamp artifact description (of a variable for a patient)	Solution
SampleTime and EnterTime differ in “year” but not in “month” field.	Replace the value of “year” field in SampleTime with that of EnterTime.
SampleTime and EnterTime differ in both “year” and “month” fields, and “year” of EnterTime is same as “year” of SampleTime next record.	Delete all records with SampleTime earlier than AdmissionTime.
SampleTime and EnterTime differ in both “year” and “month” fields, and SampleTime of current and next record only differs in “month” field, but SampleTime of current record and EnterTime of previous/next record has the same “year”.	Replace value of “year” field in SampleTime with that of previous/next record.
SampleTime and EnterTime differ in “month” not in “year” field, and absolute difference in “day” is smaller than 2 days.	Replace value of “month” field in SampleTime with that of EnterTime.
SampleTime and EnterTime differ in “month” not in “year” field, and absolute difference in “day” is <1 day, but SampleTime of current and previous/next record has same “day”.	Replace value of “day” field in SampleTime with that of previous/next record.
Absolute difference in “month” between SampleTime and EnterTime is 11, and SampleTime and EnterTime is same in “month” in previous/next record.	Add/subtract one from “year” value of SampleTime of record.
“month” and “day” of SampleTime are swapped compared to EnterTime.	Swap “month” and “day” of SampleTime according to EnterTime.
Uncorrectable SampleTime resulting in a long time gap at end of stay, and number of records after gap is less than 5 % of the total number of records.	Delete all records after gap.
Units digit of “month” of SampleTime is swapped with tens digit of “day” of SampleTime.	Swap back units digit of “month” of SampleTime with tens digit of “day” of SampleTime.
Span of SampleTime of records before gap is larger than a day while span of EnterTime of records is smaller than 6 hours, i.e. records are history long before admission.	Delete all records before AdmissionTime.
SampleTime and EnterTime only differ in tens digit of “day” field, i.e. difference in “day” is a multiple of 10.	Replace value of “day” field in SampleTime to that of EnterTime.
Date of EnterTime is the end of a month X but date of DateTime is beginning of month X-1 or X.	Replace “month” of DateTime with X+1.
SampleTime of last record before gap is before AdmissionTime (or SampleTime of first monval record for non-monval tables), and the total number of records before gaps is less than 10 % of total number of records.	Delete all records before gap.
SampleTime of first record after gap is after SampleTime of last monvals record for non-monval tables, and total number of records after gaps is less than 10 % of total number of records.	Delete all records after gap.

Supplementary Table 15: Effect of multiple horizons

Feature horizon set	AUPRC	prec (rec=0.8)	prec (rec=0.9)	prec (rec=0.95)
Short term horizon	0.555 ± 0.016	0.322 ± 0.032	0.223 ± 0.025	0.133 ± 0.041
+ Medium term horizon	0.565 ± 0.018	0.354 ± 0.029	0.226 ± 0.028	0.139 ± 0.025
+ Longer term horizon	0.568 ± 0.017	0.362 ± 0.035	0.256 ± 0.031	0.148 ± 0.024
All horizons	0.571 ± 0.019	0.363 ± 0.034	0.244 ± 0.024	0.140 ± 0.021
Only longest term horizon	0.536 ± 0.015	0.340 ± 0.037	0.225 ± 0.024	0.127 ± 0.040

Validating construction principle of multi-scale history, we check whether the addition of information for medium-term horizon, and the two longest horizons improves the prediction performance. All other features were deactivated so we can study the multi-scale history in isolation. As two baselines we have considered two schemes that do not compute multi-scale information, by just using the shortest, or the longest horizon, on its own, respectively. +/- indicates standard deviation across the n=5 independent experiments in the development splits.

Supplementary Table 16: Effect of 5-point summary

Summary function set	AUPRC	prec (rec=0.8)	prec (rec=0.9)	prec (rec=0.95)
Location	0.457 ± 0.027	0.245 ± 0.042	0.160 ± 0.030	0.105 ± 0.013
Location + Trend	0.519 ± 0.016	0.328 ± 0.030	0.220 ± 0.029	0.128 ± 0.039
All	0.570 ± 0.020	0.367 ± 0.034	0.253 ± 0.028	0.149 ± 0.020

Effect of 5-point summary just using 'location' information, vs. adding trend information (location+trend), vs 'all' possible summaries, including location/variability and trends. +/- indicates standard deviation across n=5 independent experiments in the development splits. prec: Precision, rec: Recall, AUPRC: Area under the precision recall curve.

Supplementary Table 17: Shapelet feature types

AUPRC	Random	Top	Min-Max
Count	0.36 ± 0.02	0.37 ± 0.02	0.33 ± 0.02
Min	0.40 ± 0.05	0.39 ± 0.02	0.39 ± 0.03
Distance	0.41 ± 0.05	0.41 ± 0.06	0.41 ± 0.03
Dist-set	0.41 ± 0.02	0.43 ± 0.04	0.43 ± 0.02

All possible shapelet feature extraction methods were compared by training the GBM classifier on the 5 development splits. Selection of statistically significant shapelets had little influence on the results - we therefore picked the min-max procedure because it theoretically guarantees diversity among selected shapelets. More importantly, computing features at multiple timelags (dist-set) outperformed other feature generation approaches. Therefore, we chose to extract min-max dist-set shapelet features for *circEWS*. +/- indicates standard deviation across the n=6 independent experiments performed in the temporal splits. AUPRC: Area under the precision recall curve.

Supplementary Table 18: Fixed hyperparameters for tree-based models

Parameter	Full model	Compact model	Baseline
Learning rate	0.05	0.05	0.05
Unbalanced mode	TRUE	TRUE	TRUE
Frequency of training set sub-sampling per tree	100 %	100 %	Never
Maximum depth	Log2 (Max number of leaves)	Log2 (Max number of leaves)	Log2 (Max number of leaves)
Size of subsample for histogram binning	1,000,000	1,000,000	1,000,000
Minimum #samples at leaf	1000	1000	1000
Max. #categories for cat. vars to use 1 hot encoding	100	100	100
Cat. variable (smoothing, L2)	(0.0,0.0)	(0.0,0.0)	(0.0,0.0)

Fixed hyperparameters used for tree-based (ensemble) models implemented using the *lightGBM* library. Full and compact models used the same hyperparameters, as well as the decision tree baseline, except that no random sub-sampling of the training set was used in the latter case. Cat: Categorical, L2: L2 norm of the weight vector.

Supplementary Table 19: Hyperparameter grid and optimal values for tree-based models

Parameter	Grid	Full model optimal	Compact model optimal	Baseline
Maximum number of leaves	8,16,32,64,128	128	64	128
Fraction of features randomly sampled per tree	33 %, 66 %	66 %	33 %	N/A (fixed to 100 %)
Fraction of training examples randomly sampled per tree	33 %, 66 %	66 %	66 %	N/A (fixed to 100 %)
Maximum number of trees (early stopped on validation set)	5000	142 trees (early stopped)	203 trees (early stopped)	N/A (fixed to 1)

Hyperparameter search space for the grid search for tree-based models implemented using the *lightGBM* library, yielding a grid size of 20. Decision trees had a reduced grid because feature/training set sub-sampling is not applicable for them. The last row shows the effective number of trees that are added to the ensemble, using early-stopping on the validation set, for the full/compact model respectively. N/A: Not applicable.

Supplementary Table 20: Fixed/grid hyperparameters and optimal values for LR

Parameter	Grid	Optimal
Regularization parameter	0.00001,0.0001,0.001,0.01,0.1,1.0	0.01
Loss function	N/A	Logistic
Penalty	N/A	L2
Maximum number of iterations	N/A	1000
Early stopping tolerance	N/A	1.00E-03
Learning rate schedule	N/A	Optimal
Class weight in loss function	N/A	Balanced

Hyperparameter search grid and fixed hyperparameters for the LogReg model. Only the regularization parameter weighting the L2 weight decay was included in the grid. The ‘Optimal’ learning rate schedule is based on a heuristic by Leon Bottou, and the early stopping tolerance refers to the minimal improvement on the training/validation loss that is achieved in every epoch until the optimization is early stopped.

Supplementary Table 21: Hyperparameter range and optimal values for LSTM

Parameter	Range	Optimal values
Hidden state dimension temporal	Uniform{[1000, 2000]}	1756
Hidden state dimension static	Uniform{[2, 4]}	3
Hidden state dimension joint	$\{2^\alpha : \alpha \in \text{Uniform}\{[4, 7]\}\}$	16
Learning rate	$\{10^\gamma : \gamma \in \text{Uniform}([-5, -2])\}$	1.20E-04
Dropout	Uniform([0.2, 0.8])	0.38
Activation function	{relu, tanh, sigmoid}	tanh

Random search was used to search for the optimal set of hyperparameters with the specified range for the LSTM models, and 16 sets of hyperparameters were randomly drawn. Early stopping based on the validation results was also used during the model training. Uniform(.): Uniform distribution on an interval, tanh: Hyperbolic tangent, ReLU: Rectified linear unit.